



System Verification and Validation Plan for RoCam

Team #3, SpaceY

Zifan Si

Jianqing Liu

Mike Chen

Xiaotian Lou

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Revision History

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1 Symbols, Abbreviations, and Acronyms

symbol	description
Yolo	You only look once – an existing object detection model
CNN	Convolutional Neural Network
Jetson	abbreviatio for specifcally Jetson Nano Orin, which is the proposed board for this project
STM32	abbreviation for specifically STM32 microcontroller, which is the proposed microcontroller

This Verification and Validation (V&V) plan outlines how the RoCam project will demonstrate that the system is built correctly (verification) and fits stakeholder needs (validation). It anchors testing to the SRS, architecture (MG), and detailed design (MIS), ensuring every requirement is traceable where possible. **The only testing performed is unit testing;** integration, system, and field testing are out of scope. Verification relies on structured internal reviews and checklists, plus unit tests, to confirm functional behavior and safety assumptions. Validation is limited to a planned review with the McMaster Rocketry Team (stakeholders) to collect feedback; no task-based inspection, classmate review, or supervisor validation is conducted. Roles and responsibilities are assigned to team members only, with feedback from the rocketry team captured and tracked. Priorities emphasize correctness, performance, reliability, and safety; lower-priority or out-of-scope items (e.g., exhaustive usability studies, independent verification of third-party components, and all testing beyond unit tests) are noted to keep the plan feasible. The document concludes with traceability tables, placeholders for unit-level details after design finalization, and appendices for parameters and optional surveys—forming a clear, practical roadmap for the evidence of quality that can be gathered under these constraints.

2 General Information

2.1 Summary

The RoCam system is a ground-based camera and gimbal assembly designed to autonomously track model rockets during launch and flight. It utilizes a high-resolution camera coupled with a motorized gimbal to maintain visual lock on the rocket, providing real-time data and video feed. The system consists of a STM32 based motion controller that interfaces with the physical gimbal; a Jetson compute module that processes the camera feed; and a react frontend that displays the video preview and allows the user to control the gimbal. The primary purpose of RoCam is to replace the traditional unreliable manual tracking methods with an automated solution to produce high-quality flight data and video footage. To better assist model rocket teams across Canada to conduct successful launches by providing an affordable and reliable system that can be easily track launching movements.

2.2 Objectives

The primary objective of this Verification and Validation V&V Plan is to establish confidence in the correctness, performance, and reliability of the RoCam system by ensuring that all verification and validation activities are thorough, traceable, and aligned with project requirements. The plan emphasizes unit testing and internal review to confirm that functional and performance criteria are addressed where possible, with improvements tracked through GitHub Issues. Validation focuses on a planned review with the McMaster Rocketry Team (stakeholders) to collect feedback; no field testing or supervisor validation is performed. These qualities—correctness, performance, reliability, and traceability—are fundamental to the success of the V&V process, ensuring that the RoCam system is both technically sound and operationally dependable. Certain objectives, such as independent verification of third-

party hardware and extended usability testing, are excluded due to scope and resource limitations, with reliance placed on manufacturer and stakeholder validation.

2.3 In-Scope Objectives

Verification and validation efforts will prioritize the following qualities:

- **Correctness:** Ensure the reliability of tracking and control algorithms through testing and validation.
- **Performance:** Achieve real-time processing of 1080p 60 fps video and maintain minimal gimbal control latency.
- **Safety:** Guarantee that the system operates safely under assumed conditions, with fail-safe mechanisms.

2.4 Out-of-Scope Objectives

The following objectives are excluded from this project due to time, budget, and resource constraints:

- **Third-Party Component Verification:** External hardware such as the gimbal controller and camera are assumed to perform according to their manufacturer specifications.
- **Scalability Beyond Small-Scale Rockets:** Tracking a bullet or extremely difficult targets beyond a model rocket is outside the scope of this project.

By defining these priorities and exclusions, the RoCam project ensures that available effort is directed toward validating the most critical system qualities: **performance, accuracy, and operational safety.**

2.5 Extras

Extras

Beyond the core objectives, the RoCam project includes several extras that enhance its quality and usability:

- **STM32 control circuit design:** Development of a custom control circuit to interface the gimbal actuators with the host computer, enabling precise motion control and feedback sensing.
- **User Instructional Video:** Creation of a short instructional video demonstrating system setup, calibration, and operation to support new users and ensure safe deployment.

These extras aim to improve the overall usability, accessibility, and documentation quality of the final deliverable, reinforcing RoCam's emphasis on practical integration between computer vision software, mechanical control, and end-user experience.

2.6 Relevant Documentation

Software Requirements Specification (SRS Vision-Guided Tracker Team (2025c)):

The SRS document serves as an outline for the VnV plan. The SRS document contains specific functional and non-functional requirements, performance criteria, and constraints. The SRS document also contains detailed stakeholder roles and scope of the work. Aligning the VnV plan with the SRS document ensures that all specified requirements are adequately verified and validated.

Software Architecture MG (MG Vision-Guided Tracker Team (2025a)) The MG document provides details of our system architecture. To effectively design tests to validate and verify the system, it is important to understand the architecture and functionality of each module.

Software Detailed Design Document (MIS Vision-Guided Tracker Team (2025b)):

The MIS document provides detailed design of our system. It provides critical information about the implementation details, including data structures, algorithms, and interfaces. This document is essential for efficient test design to specific modules.

3 Plan

This section outlines the overall Verification and Validation (V&V) plan for the **RoCam** project. It defines the structure, responsibilities, and methods used to verify that the system meets its functional and non-functional requirements and to validate that it fulfills stakeholder needs. The section covers the verification of the SRS, design, and implementation via internal reviews and unit testing only, as well as the validation strategy through a planned stakeholder review with the McMaster Rocketry Team (no field testing). Its purpose is to provide a clear, organized framework ensuring that all aspects of RoCam are systematically tested, reviewed, and proven reliable before final deployment.

3.1 Verification and Validation Team

Table 1: Verification and Validation Team Responsibilities

Name	Role	V&V Responsibilities
Zifan Si	Front End Interface Tester	Responsible for verifying and validating the usability and functionality of the react frontend interface. Ensure usability, robustness of frontend features.
Jianqing Liu	Team Lead	Conduct integration test and system validation. Validate the functionality and performance of the system against requirements and use cases. Validate system design and functionality.
Mike Chen	CV Tester	Verify and validate computer vision pipeline performance, including object detection and tracking accuracy.
Xiaotian Lou	Integration Tester and test automation developer	Implement automated testing frameworks in all software components. Implement API, and integration test across different modules.

3.2 SRS Verification

The SRS verification will focus on verify whether the system design satisfy all functional and non-functional requirements, stakeholder needs, scope of work and constraints specified. The SRS verification will be conducted through the following steps:

3.2.1 functional and Non-Functional Requirement Review

Since the Design function will evaluate in detail whether the system align with the functional and non-functional requirements. The SRS verification will focus on verify the system design as a whole against the requirements. It will be implemented as integration test in the final stages of development.

3.2.2 Stakeholder Review

Since our team leader is the control team lead of the McMaster Rocketry Team, He will be the one to review the system design and implementation and the communication bridge between the team and the stakeholders.

3.2.3 Constraints and Scope of Work Review

Several internal review sessions will be conducted to ensure that our system design and implementation are within the defined scope of work and constraints specified in the SRS document.

Table 2: SRS Verification Checklist

Phase	Verification Activities
Functional & Non-Functional Requirement Review	☐ Review all requirements against current system design ☐ Create automated system tests for functional requirements ☐ Measure performance, safety, and reliability against non-functional requirements ☐ Record test results and address any failed cases ☐ Update traceability matrix after verification
Stakeholder Review	☐ Present verified results and collect feedback ☐ Document required changes based on feedback ☐ Obtain stakeholder confirmation or approval
Constraints & Scope of Work Review	☐ Verify implementation stays within project scope ☐ Check compliance with all technical and safety constraints ☐ Review design against timeline and budget limits ☐ Update records if scope or constraints change
Integration Verification	☐ Conduct integration tests covering all system components ☐ Validate full system behavior against SRS requirements ☐ Log and resolve all integration issues ☐ Prepare final verification report for stakeholder review
Safety Review	☐ Review safety requirements against Hazard Analysis

3.3 Design Verification

The design verification process for the **RoCam** project ensures that the proposed architecture, algorithms, and hardware interfaces satisfy all functional and non-functional requirements before implementation. Design verification activities will be conducted through structured design reviews, peer evaluations, and the use of detailed verification checklists. The goal is to confirm that the design meets the intended performance, reliability, and safety objectives defined in the SRS.

Verification Plan

The design verification will proceed through internal review only (no classmate or supervisor review, due to contact and resource limitations):

1. **Internal Design Review:** Each subsystem (computer vision, control, and UI) will undergo an internal review by each team member. Each member must agree on the designed architecture, algorithms, and interfaces. Interfaces must be clearly defined for scalability.

Verification Artifacts

Design verification will produce the following artifacts:

- Verified design diagrams and interface specifications.
- Internal review notes and agreed-upon design decisions.
- Updated design documents reflecting corrections and improvements.
- Completed design verification checklists (see below).

Design Verification Checklist

The following checklist will be used during design reviews to ensure completeness and consistency across all modules:

Table 3: Design Verification Checklist

Phase	Verification Activities
Internal Design Review	☐ Review subsystem architectures (vision, control, UI) with all team members ☐ Confirm APIs that are agreed upon ☐ Verify all module interfaces are clearly defined and documented in code ☐ Ensure design supports scalability and modularity ☐ Record review comments and update design diagrams
Verification Artifacts	☐ Store verified design diagrams and updated interface specifications ☐ Archive internal review notes and design decisions ☐ Maintain a design document in the project repository ☐ Complete and file the final design verification checklist

By following this structured verification plan and checklist, the team ensures that the RoCam design is both technically sound and fully aligned with its performance and safety goals prior to implementation.

3.4 Verification and Validation Plan Verification

The verification of the verification and validation (VnV) plan for the **RoCam** project will be verified through internal review and unit testing only. Each iteration of feedback and improvements will be tracked with GitHub Issues. The focus of the verification plan is to ensure functionality and performance metrics are adequately covered in design where possible.

For the validation plan, the only validation activity is a planned review with the McMaster Rocketry Team (stakeholders) to collect feedback. No field tests, integration tests, or supervisor validation are performed; the only testing conducted is unit testing.

Table 4: RoCam Verification and Validation (V&V) Checklist

Phase	Actionable Verification and Validation Activities
Verification Plan Review	☐ Conduct internal reviews of the V&V plan with team ☐ Ensure functionality and performance metrics are fully covered ☐ Track feedback and revisions using GitHub Issues ☐ Confirm all review comments are resolved before approval
Automated Verification Testing	☐ Develop automated tests for vision, control, and UI components ☐ Verify software behavior under simulated conditions ☐ Record test results and compare against performance criteria ☐ Update test scripts based on identified defects or regressions
Continuous Monitoring	☐ Review unit test outcomes after each iteration ☐ Maintain test evidence and validation summaries in repository ☐ Update the V&V plan to reflect system improvements and new unit test cases

3.5 Implementation Verification

The implementation verification for the **RoCam** project ensures that the developed software and integrated hardware components correctly realize the verified design. This phase confirms that the implemented system satisfies the functional and non-functional requirements defined in the SRS through a combination of structured testing, static verification, and peer review.

Dynamic Verification

Due to resource and contact limitations, **only unit testing** is performed. Integration testing, system testing, and field testing are out of scope.

- **Unit Testing:** Each software module (vision processing, gimbal control, data recording, and UI management) will undergo unit testing to verify functional correctness in isolation. This is the only form of dynamic testing conducted.

Out of scope: Integration testing, system testing, and field testing are not performed due to the constraints noted in this plan.

Static Verification

Static verification will be performed throughout the implementation phase to ensure that the codebase meets quality, safety, and maintainability standards before execution. Planned techniques include:

- **Code Walkthroughs:** Internal code walkthroughs where each subsystem lead presents core implementation logic and design decisions to the team, enabling identification of potential design or coding issues.
- **Code Inspection:** Team members will review each other's pull requests prior to merging into the main branch. Reviews will focus on coding standard compliance, potential logic errors, exception handling, and safety-critical sections align with the Hazard Analysis (e.g., gimbal actuation and state transitions).
- **Static Analysis Tools:** Where applicable (e.g., on the frontend), static analysis may be used to detect common implementation defects. Results will be logged and reviewed as part of the verification record.
- **Documentation Review:** The team will verify that all inline comments, module-level documentation, and API references are consistent with the design specification and user documentation requirements.

Together, these verification activities ensure that the RoCam implementation is thoroughly tested, statically verified, and validated against both its design and real-world performance expectations.

3.6 Automated Testing and Verification Tools

The RoCam project adopts a structured automated testing approach that prioritizes maintainability, integration compatibility, and developer productivity. The selection of testing frameworks was guided by three key criteria: (1) **language and runtime fit**, ensuring each tool aligns with its respective language and runtime; (2) **ease of automation**, to integrate smoothly with CI/CD pipelines; and (3) **scalability**, so that test coverage can expand as the project evolves. These principles ensure a consistent and reliable testing environment across frontend and end-to-end validation stages.

Each subsystem of RoCam employs unit testing and coverage tools appropriate to its implementation language. The frontend (React/TypeScript) uses a unit testing framework and coverage reporting suited to that stack. System-level and Web UI testing, where in scope, use tooling that can coordinate frontend and hardware layers. Specific frameworks are chosen to support CI/CD integration and to allow test coverage to scale as the project evolves.

Code Coverage and Reporting Plan:

Coverage analysis is conducted using tools that provide line, branch, and function-level metrics for each language stack, with results suitable for unified tracking and CI integration.

Linters and Formatters:

To maintain coding standards and prevent stylistic or logical errors, RoCam enforces consistent linting and formatting on the frontend:

- **Frontend (TypeScript):** Uses **ESLint** and **Prettier** to maintain uniform style rules and detect syntax or logical inconsistencies.

3.7 Software Validation

Software validation confirms that **RoCam** satisfies stakeholder needs and accurately implements the intended use cases (“building the right system”). Validation complements verification and is referenced to the SRS goals and stakeholder roles.

Stakeholder Review (McMaster Rocketry Team)

Validation is limited to a planned review with the McMaster Rocketry Team (stakeholders). The following is **in scope**:

- **Stakeholder Review Session:** A scheduled review with the McMaster Rocketry Team to present the current system (or design/SRS), discuss scope, priorities, and acceptance criteria, and collect feedback. Revisions to the plan or implementation will be made based on their feedback.

Out of scope (not conducted): Task-based inspection by stakeholders, classmate or peer review, supervisor validation, Rev 0/supervisor demo, and formal user (operator) sessions. These are excluded due to contact and resource limitations.

External Data and Bench Validation

- **Reference Footage:** Where live launches are unavailable, public model-rocket videos and internally recorded dry-runs are used to validate detection/tracking logic and reacquisition behavior.
- **Synthetic Scenarios:** Scripted clips with occlusion/smoke/backlight and varying apparent rocket sizes validate robustness claims prior to field tests.

Acceptance and Traceability

- **Acceptance Criteria:** Derived from the SRS (e.g., sustained 1080p 60 fps I/O, end-to-end latency ≤ 120 ms, successful tracking through nominal flight phases, recording with a ngle overlays).
- **Traceability:** Validation tests map to SRS goals and stakeholder needs; a table links each validation activity to the corresponding SRS item(s) and planned evidence (videos, logs, checklists).

If no suitable external datasets are available for certain scenarios, this limitation will be stated explicitly; validation in this plan relies on the planned stakeholder review with the McMaster Rocketry Team and, where applicable, unit testing and internal review only. This section also references the SRS verification section for consistency checks between validated behaviors and specified requirements.

4 System Tests

Scope note: Due to resource and contact limitations, the only testing actually performed in this project is **unit testing**. The system tests and field tests described in this section are documented for reference and traceability but are **out of scope** and will not be executed. Evidence of correctness will come from unit tests and from the planned stakeholder review with the McMaster Rocketry Team.

4.1 Tests for Functional Requirements

The following manual tests are designed to evaluate the robustness of the Tracking Camera System. Each test includes its initial state, input, expected output, justification (test case derivation), and procedure. Tests are organized by functional area. The intent is to verify compliance with the functional requirements (FR) while deliberately stressing the system to expose edge cases.

4.1.1 Area of Testing: Video Acquisition and Output

test-acquire-frame: Camera Frames are Acquired at Startup

- **Control:** Manual
- **Initial State:** System powered off; camera connected; HDMI monitor attached.
- **Input:** Power on the system and allow it to fully initialize.
- **Expected Output:** Within 10s of boot, logs show that the camera is acquiring frames.
- **Test Case Derivation:** The system's operation depends on a valid video feed.
- **How Test Will Be Performed:** Boot the device; observe the program logs.

test-disconnect-camera: Camera Disconnect During Operation

- **Control:** Manual
- **Initial State:** System initialized; preview and HDMI output active.
- **Input:** Physically disconnect the camera from the system.
- **Expected Output:** On disconnect, the UI and HDMI output show a clear error message; no crash.
- **Test Case Derivation:** When the camera is disconnected, the system should gracefully handle the loss without crashing. And it should notify the operator that the camera is disconnected.
- **How Test Will Be Performed:** Disconnect the camera from the system; observe the UI and HDMI output for error messages.

test-hdmi-output: HDMI Output with Info Overlay

- **Control:** Manual
- **Initial State:** System initialized; idle state; preview and HDMI output active.
- **Input:** Move the gimbal manually through the UI.
- **Expected Output:** HDMI shows 1080p60 camera feed with a text overlay of current pan/tilt angles.
- **Test Case Derivation:** The system should be able to display the current camera feed and the pan/tilt angles for the viewers
- **How Test Will Be Performed:** Move the gimbal manually, and visually verify overlay values change smoothly and match commanded motion

test-disconnect-hdmi: HDMI Disconnect

- **Control:** Manual
- **Initial State:** System initialized; preview and HDMI output active.
- **Input:** Physically disconnect the HDMI cable.
- **Expected Output:** On disconnect, the UI show a clear error message; no crash.
- **Test Case Derivation:** When the HDMI cable is disconnected, the system should gracefully handle the loss without crashing. And it should notify the operator that the HDMI cable is disconnected.
- **How Test Will Be Performed:** Disconnect the HDMI cable from the system; observe the UI for error messages.

test-preview-real-time: Preview with Real-Time Info

- **Control:** Manual
- **Initial State:** System initialized; idle state; preview and HDMI output active.
- **Input:** Move the gimbal manually through the UI.
- **Expected Output:** Preview in the UI updates fluidly (target ≥ 15 fps) and displays current pan/tilt angles.
- **Test Case Derivation:** The system should allow the camera operator to see the current pan/tilt angles and HDMI output in real-time.
- **How Test Will Be Performed:** Move the gimbal manually, and visually verify the preview in the UI updates fluidly and displays the current pan/tilt angles.

test-network-disconnect: Network Disconnect While Viewing UI

- **Control:** Manual
- **Initial State:** System initialized; preview and HDMI output active.
- **Input:** Disconnect the system from the network for 30s; reconnect.
- **Expected Output:** The system keeps functioning as normal without network; UI show a clear error message; upon reconnection, the UI session restores automatically within 5s without reboot; no data loss or crash.
- **Test Case Derivation:** The system need to be resilient against transient network failures in field conditions.
- **How Test Will Be Performed:** Manually disconnect and connect the system from the network; observe the UI for error messages.

4.1.2 Area of Testing: Recording

test-recording: Start and Stop Recording

- **Control:** Manual
- **Initial State:** System initialized; idle state; sufficient disk space.
- **Input:** Press "Start Recording", wait 10s, then press "Stop Recording".
- **Expected Output:** A 10 second 1080p60 video file is created and saved to the disk.
- **Test Case Derivation:** The system should be able to start and stop recording.
- **How Test Will Be Performed:** Press "Start Recording", wait 10s, then press "Stop Recording".

test-disk-full: Disk Full Behavior During Recording

- **Control:** Manual
- **Initial State:** System initialized; idle state; storage nearly full (simulate by filling disk leaving < 1min recording capacity).
- **Input:** Start recording and allow disk to fill.
- **Expected Output:** UI show a clear "Storage Full" alert; current recording is stopped and the file is saved to the disk; no crash.
- **Test Case Derivation:** The system should be able to handle storage exhaustion gracefully.
- **How Test Will Be Performed:** Fill the disk with data until it is nearly full; start recording; observe the UI for the "Storage Full" alert; check the disk for the saved file.

test-manage-recordings: List and Download Recordings

- **Control:** Manual
- **Initial State:** At least two prior recordings exist.
- **Input:** Open recordings list; download the latest file and an older file; delete a small test clip.
- **Expected Output:** List shows correct entries (name, timestamp, duration); downloads succeed; delete removes the selected item only.
- **Test Case Derivation:** The system should be able to list, delete, and download recordings.
- **How Test Will Be Performed:** Open the recordings list; download the latest file and an older file; delete a small test clip; check the list for the correct entries; check the disk for the saved files.

4.1.3 Area of Testing: Tracking

test-manual-control: Manual Gimbal Control in Idle

- **Control:** Manual
- **Initial State:** System initialized; idle state; preview and HDMI output active.
- **Input:** Issue manual pan and tilt commands via UI.
- **Expected Output:** The gimbal follows commands smoothly.
- **Test Case Derivation:** The operator should be able to manually control the gimbal in idle state.
- **How Test Will Be Performed:** Issue manual pan and tilt commands via UI; observe the gimbal movement.

test-transition-to-tracking: Transition to Tracking When Rocket Detected

- **Control:** Manual
- **Initial State:** System initialized; idle state; preview and HDMI output active.
- **Input:** Arm the system via UI, and waving a model rocket in front of the camera.
- **Expected Output:** The system transitions to tracking state as soon as the rocket is detected; the UI shows the system is in tracking state
- **Test Case Derivation:** The system should be able to transition to tracking state when a rocket is detected.
- **How Test Will Be Performed:** Arm the system via UI, and waving a model rocket in front of the camera; observe the UI for the system to transition to tracking state.

test-tracking: Keep Rocket in Frame While Tracking

- **Control:** Manual
- **Initial State:** System initialized; tracking state; preview and HDMI output active.
- **Input:** Continue waving the model rocket in front of the camera.
- **Expected Output:** Gimbal should continuously adjust its position to keep the rocket in the frame for the duration of the test.
- **Test Case Derivation:** The system should be able to keep the rocket in the frame while tracking.
- **How Test Will Be Performed:** Wave the model rocket in front of the camera; observe the gimbal movement; observe the preview in the UI.

test-return-to-idle: Return to Idle When Rocket is Lost

- **Control:** Manual
- **Initial State:** System initialized; tracking state; preview and HDMI output active.
- **Input:** Occlude the model rocket.
- **Expected Output:** Within 2s of sustained loss, system transitions to idle state; ceases autonomous gimbal commands; UI indicates "target lost".
- **Test Case Derivation:** The system should be able to return to idle state when the rocket is lost.
- **How Test Will Be Performed:** Occlude the model rocket; observe the UI for the system to transition to idle state.

test-exit-gimbal-range: Rocket Exits Gimbal Range

- **Control:** Manual
- **Initial State:** System initialized; tracking state; preview and HDMI output active.
- **Input:** Move the model rocket outside of the gimbal's range of motion.
- **Expected Output:** The gimbal moves until it reaches its mechanical limit, then stops moving further; no errors or warnings are presented to the user; system remains responsive.
- **Test Case Derivation:** The system must safely handle objects that leave the gimbal's range without faulting or reporting spurious errors.
- **How Test Will Be Performed:** Move the model rocket outside of the gimbal's range of motion; observe the gimbal reaches its mechanical end and stops; verify that no error messages appear in the UI and that all other system functions remain operational.

4.1.4 Field Testing (Out of Scope)

Field testing is **out of scope** for this project due to resource and contact limitations. The following is retained for reference only.

If field testing were to be conducted, it would be done by the McMaster Rocketry Team (client), who would be provided with the manual and would operate the system at a model rocket launch event. The development team would be present to assist. After the event, the client could fill out a survey (Appendix 6.2). Under the current plan, no field testing is performed; only unit testing and a planned stakeholder review with the rocketry team are conducted.

4.2 Tests for Nonfunctional Requirements

4.2.1 UI Walkthrough

The UI walkthrough will be conducted by the capstone team members themselves before the field test.

test-consistent-units: Consistent and Changeable Units

- **Type:** Manual
- **Initial State:** System initialized; idle state; preview and HDMI output active.
- **Input/Condition:** Go through the UI twice, once with metric units, once with imperial units.
- **Output/Result:** The units are consistent and changeable.
- **How Test Will Be Performed:** The capstone team members will go through the UI twice, once with metric units, once with imperial units; observe the units in the UI.

test-bilingual-ui: Bilingual UI

- **Type:** Manual
- **Initial State:** System initialized; language = English.
- **Input/Condition:** Switch UI language to French and back; navigate all UI sections, dialogs, and toasts.
- **Output/Result:** All visible strings/localized assets appear in the selected language; no truncation/overflow; no mixed-language strings; hotkeys remain functional; date/time/number formats localize correctly.
- **How Test Will Be Performed:** Perform a complete UI walkthrough in French; capture screenshots of each page/dialog; switch back to English and repeat spot checks.

test-immediate-ui-feedback: Immediate UI Feedback

- **Type:** Manual
- **Initial State:** System idle; UI ready.
- **Input/Condition:** Trigger representative actions: button press, toggle, menu selection, starting/stopping recording, arming/disarming, opening settings.
- **Output/Result:** A visible confirmation (pressed state, spinner, toast, label change) appears within 0.2s for each action.
- **How Test Will Be Performed:** Observe the UI while performing each action

test-visual-indicators: Rely on Visual Indicators

- **Type:** Manual
- **Initial State:** System initialized; speakers muted or disconnected.
- **Input/Condition:** Trigger statuses/alerts: armed, tracking, target lost, storage low/full, network disconnect, device error.
- **Output/Result:** Every status/alert is presented visually (icons, colors, banners, toasts) without relying on sound; no action-critical info is audio-only.
- **How Test Will Be Performed:** With audio disabled, induce each state/fault; verify a visible indicator appears and remains long enough to be noticed.

test-confirmation-prompts: Confirmation Prompts for Safety-Critical Actions

- **Type:** Manual
- **Initial State:** System idle; gimbal powered.
- **Input/Condition:** Attempt potentially hazardous actions: force home while tracking, hard stop, high-speed slew, firmware update during armed state, factory reset.
- **Output/Result:** A blocking confirmation dialog appears for each action; dialog text states the specific risk and outcome; default action is “Cancel”; action proceeds only after explicit confirmation.
- **How Test Will Be Performed:** Trigger each action; capture screenshots of the prompt; verify risk wording is specific (not generic); verify no motion/change occurs until “Confirm” is pressed.

test-manual-idle-mode: Manual Idle Mode / Override

- **Type:** Manual
- **Initial State:** System armed or tracking.
- **Input/Condition:** Press “Idle” (or equivalent) control while tracking.
- **Output/Result:** System transitions to idle within 1 s; autonomous commands cease; manual controls become available; status clearly shows “Idle.”
- **How Test Will Be Performed:** Enter tracking, then invoke manual idle; observe immediate cessation of gimbal motion; verify state banner change and manual jog controls responsiveness.

test-no-collection-pii: No Collection of PII

- **Type:** Manual
- **Initial State:** System initialized; default configuration.
- **Input/Condition:** Inspect all settings, logs, exported files, filenames/paths, and network-configurable fields.
- **Output/Result:** No fields request or store personal identifiers (names, emails, phone numbers, precise home addresses, faces with identity tags); logs contain only technical data; telemetry excludes PII.
- **How Test Will Be Performed:** Open each settings and log panel; download logs/recordings; inspect metadata (EXIF/JSON/CSV) for PII keys; verify any optional labels are device/session IDs, not personal data.

test-color-vision-deficiency: Color-Vision Deficiency Accessibility

- **Type:** Manual
- **Initial State:** System initialized; all status indicators visible.
- **Input/Condition:** Review all status/alert states (normal, armed, tracking, warning, error) and interactive controls (enabled/disabled) under simulated Protanopia/Deuteranopia/Tritanopia or printed CVD reference swatches.
- **Output/Result:** Each state is distinguishable without relying on hue alone (uses shape, text labels, patterns, or contrast); contrast ratio $\geq 4.5:1$ for text/icons against background.
- **How Test Will Be Performed:** Cycle UI through all states; verify distinct icon shapes/labels; optionally view via a CVD simulator; record pass/fail with screenshots; confirm tooltips/textual labels exist for each color state.

4.2.2 Code and Documentation Walkthrough

test-manual-arm-mode: Manual Arm Mode Only

- **Type:** Static
- **Initial State:** N/A
- **Input/Condition:** Inspect all state transition code and configuration defaults.
- **Output/Result:** The system only enters armed mode when explicitly commanded by the user, never automatically.
- **How Test Will Be Performed:** Review code for all references to the armed state, verify guards require user input.

test-gimbal-interface: Gimbal Interface Documentation Review

- **Type:** Static
- **Initial State:** N/A
- **Input/Condition:** Review the interface specification with the client, including command set, telemetry, timing, and safety notes.
- **Output/Result:** The client confirms the interface documentation is complete, clear, and suitable for supporting multiple gimbal systems.
- **How Test Will Be Performed:** Conduct a walkthrough meeting with the client, check all required topics against a prepared checklist.

4.2.3 Error Handling and Logging

test-report-errors: Report All Errors to the User

- **Type:** Manual
- **Initial State:** System initialized and idle.
- **Input/Condition:** Manually trigger hardware, communication, and software faults.
- **Output/Result:** Each error generates a visible and descriptive message on the user interface.
- **How Test Will Be Performed:** Disconnect devices, kill processes, and simulate faults while observing the UI for clear error reporting.

test-stop-immediately: Stop Immediately on Unrecoverable Error

- **Type:** Manual
- **Initial State:** System operating in tracking mode.
- **Input/Condition:** Introduce a fatal condition such as gimbal communication loss or corrupted control process.
- **Output/Result:** The system immediately stops all automated operations and transitions to a safe idle state.
- **How Test Will Be Performed:** Disconnect the gimbal or crash the tracking process, verify that motion stops and the system displays a clear unrecoverable error message.

test-validate-commands: Validate All Commands

- **Type:** Manual
- **Initial State:** System idle.
- **Input/Condition:** Enter invalid or out-of-range user commands through the UI or API.
- **Output/Result:** The system rejects invalid commands and displays a validation error without sending them to the gimbal.
- **How Test Will Be Performed:** Attempt to send commands beyond gimbal limits or in unsupported formats and confirm that they are blocked with clear feedback.

test-log-operations: Log All Operations

- **Type:** Manual
- **Initial State:** System initialized and user logged in.
- **Input/Condition:** Perform a series of user operations such as arming, tracking, and recording.
- **Output/Result:** Each action is logged with a timestamp and operation details in the system log.
- **How Test Will Be Performed:** After the field test, review the log file to confirm all actions and timestamps are recorded accurately.

4.2.4 Field Testing (Out of Scope)

Field testing and user surveys are out of scope; only unit testing and the stakeholder review are performed.

test-user-survey: User Survey (Out of Scope)

- **Type:** Static (out of scope)
- **Initial State:** N/A
- **Input/Condition:** N/A
- **Output/Result:** Completed user survey.
- **How Test Will Be Performed:** Not performed. If field testing were in scope, the client would fill out the survey after the field test. Under the current plan, only unit testing and stakeholder review are conducted.

4.3 Traceability Between Test Cases and Requirements

Test ID	Requirements
test-acquire-frame	FR-1, SLR-2
test-disconnect-camera	FR-1
test-hdmi-output	FR-8, FR-9, INT-1, SLR-3
test-disconnect-hdmi	FR-8
test-preview-real-time	FR-10
test-network-disconnect	FR-10
test-recording	FR-11
test-disk-full	FR-11
test-manage-recordings	FR-12
test-manual-control	FR-2, FR-6
test-transition-to-tracking	FR-3, FR-4
test-tracking	FR-2, FR-3, FR-7
test-return-to-idle	FR-5
test-exit-gimbal-range	FR-2, FR-7
test-consistent-units	EZ-1, PI-1
test-bilingual-ui	PI-2
test-immediate-ui-feedback	EZ-2
test-visual-indicators	EZ-3
test-confirmation-prompts	EZ-4, UPR-3
test-manual-idle-mode	SCR-1
test-no-collection-pii	LR-1
test-color-vision-deficiency	AR-1
test-manual-arm-mode	SCR-4
test-gimbal-interface	INT-2
test-report-errors	RFR-1
test-stop-immediately	RFR-2
test-validate-commands	IR-1
test-log-operations	AUR-1
test-user-survey	AR-1, AR-2, SR-1, LR-1, UPR-1, UPR-2, PAR-1, SLR-1, RFR-5, CR-1, CR-2, SCR-2, SCR-3, RFR-3, EPE-1

5 Unit Test Description

5.1 Unit Testing Scope

5.2 Tests for Functional Requirements

5.2.1 Module 1

1. test-id1

Type:

Initial State:

Input:

Output:

Test Case Derivation:

How test will be performed:

2. test-id2

Type:

Initial State:

Input:

Output:

Test Case Derivation:

How test will be performed:

3. ...

5.2.2 Module 2

...

5.3 Tests for Nonfunctional Requirements

5.3.1 Module ?

1. test-id1

Type:

Initial State:

Input/Condition:

Output/Result:

How test will be performed:

2. test-id2

Type: Functional, Dynamic, Manual, Static etc.

Initial State:

Input:

Output:

How test will be performed:

5.3.2 Module ?

...

5.4 Traceability Between Test Cases and Modules

References

Vision-Guided Tracker Team. Software architecture (module guide), 2025a. Project document, this repository.

Vision-Guided Tracker Team. Software detailed design (module interface specification), 2025b. Project document, this repository.

Vision-Guided Tracker Team. Software requirements specification, 2025c. Project document, this repository.

6 Appendix

This is where you can place additional information.

6.1 Symbolic Parameters

The definition of the test cases will call for SYMBOLIC_CONSTANTS. Their values are defined in this section for easy maintenance.

6.2 User Survey Questions

Please answer each question based on your experience during the field test. For questions rated on a scale, circle the number that best reflects your opinion: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree.

User Interface

1. The interface was readable and easy to see outdoors. (1 2 3 4 5)
2. The interface looked professional and well-designed. (1 2 3 4 5)
3. The important information was immediately visible on the screen. (1 2 3 4 5)
4. No unnecessary or confusing information was shown. (1 2 3 4 5)
5. The system was easy to learn and operate after a short time. (1 2 3 4 5)
6. The terms and labels used in the interface were clear and understandable. (1 2 3 4 5)

System Performance

1. The tracking footage was stable and smooth. (1 2 3 4 5)
2. The tracking system maintained lock on the rocket reliably. (1 2 3 4 5)
3. The system operated as expected without major errors. (1 2 3 4 5)
4. Once armed, the system operated fully autonomously. (1 2 3 4 5)
5. I was able to fully operate and monitor the system remotely. (1 2 3 4 5)

User Experience and Feedback

1. Were any colors, symbols, or content elements disrespectful or inappropriate to you?
(Please describe if applicable)

2. What improvements would you suggest for future versions of RoCam?

Appendix — Reflection

The information in this section will be used to evaluate the team members on the graduate attribute of Lifelong Learning.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. What knowledge and skills will the team collectively need to acquire to successfully complete the verification and validation of your project? Examples of possible knowledge and skills include dynamic testing knowledge, static testing knowledge, specific tool usage, Valgrind etc. You should look to identify at least one item for each team member.
4. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?

Jianqing Liu

1. **What went well:** I was responsible for the Part 4 of this deliverable, focusing on all the system tests. I think it was really easy to come up with all the test cases because We already completed the SRS.
2. **Pain points and resolution:** I think some of the test cases are very verbose. For some of the tests, how test will be performed is just the same as the input and output condition of the test case.

Mike Chen

1. **What went well:** I contributed primarily to Part 3 of this deliverable, focusing on the verification and validation plan and the automation tools section. The structure

and clarity of this part improved substantially after integrating team and stakeholder feedback. I ensured that each verification activity was directly tied to the SRS and our test framework.

2. **Pain points and resolution:** I initially faced several issues with GitHub CI configuration—some commits overwrote teammates’ work and caused LaTeX compilation failures. I resolved these by restoring the affected commits, redesigning the YAML workflows, and setting up branch protection rules. Revisions based on feedback also required extensive reformatting to achieve consistency across sections.
3. **Knowledge and skills to acquire:** I plan to improve my skills in CI/CD automation and verification planning, particularly integrating automated builds, test execution, and documentation generation through GitHub Actions.
4. **Approaches to acquire skills:** I will (1) study advanced GitHub Actions pipelines from open-source projects and (2) create a smaller test repository to simulate multi-stage verification workflows. I selected the second approach because it provides hands-on experience with merge control, build automation, and verification feedback integration.

Frank

1. **What went well:** I contributed to both Part 2 and Part 3 of the deliverable, ensuring that the design and verification sections were fully aligned with the SRS. My work focused on refining the test structure, verifying requirement traceability, and improving the flow between verification items and their corresponding system requirements.
2. **Pain points and resolution:** The main challenge was maintaining consistent terminology and depth of explanation between sections written by different team members. To resolve this, I reviewed the SRS and cross-checked every requirement reference to ensure correctness and coherence across parts.
3. **Knowledge and skills to acquire:** I plan to strengthen my knowledge of requirements traceability and automated testing frameworks to improve coverage and maintain consistency with system goals.
4. **Approaches to acquire skills:** I will (1) review documentation on structured traceability mapping and (2) develop example UI and integration tests that link directly to specific SRS requirements. I chose the second approach because applying traceability in practice will reinforce both documentation and test alignment.

Xiaotian Lou

1. **What went well:** I focused mainly on Part 2 of the deliverable, organizing the SRS and design verification sections. The verification checklists and phase structure improved document readability and made the review process more systematic.

2. **Pain points and resolution:** The primary difficulty was harmonizing writing styles and section formatting across contributors. I standardized tables, checklists, and layout conventions to produce a consistent and professional presentation.
3. **Knowledge and skills to acquire:** I aim to gain more experience in formal verification documentation and structured validation reporting to improve the traceability and completeness of future deliverables.
4. **Approaches to acquire skills:** I will (1) study professional verification and validation templates from engineering projects and (2) apply checklist-based verification methods in upcoming development stages. I selected the first approach to strengthen my understanding of documentation standards and best practices.